Will Dienstfrey

Ph.D. Student in Earth and Space Sciences | University of Washington

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Research Interests

Fiber-optic sensing; observational seismology; cryosphere geophysics; seismic tomography; inverse problems; physics-based modeling of Earth systems.

Education

University of Washington 2025–Present
Ph.D. Student, Department of Earth and Space Sciences
Advisor: Prof. Brad Lipovsky

Amherst College 2020–2024
B.A. Physics, *Magna cum laude*
Honors Thesis Advisor: Prof. Nicholas Holschuh

Research Experience

Graduate Researcher 2025–Present
University of Washington *Seattle, WA*

- Developing distributed acoustic sensing (DAS) methods to investigate the structure and dynamics of marine, volcanic, and glacial environments.
- Build computational workflows for analyzing ambient noise and natural source seismic data and signal processing, surface wave imaging, and time-lapse monitoring.

Seismic Data Analyst Jul. 2024–Aug 2025
Alaska Earthquake Center *Fairbanks, AK*

- Located earthquakes and calculate magnitudes using GENLOC and regional seismic-network data.
- Contributed to weekly and monthly seismicity reports used by researchers, state and federal agencies, and the public.
- Serviced seismic stations in remote field settings across Alaska.

Gregory S. Call Research Fellow Jun. 2021–Jun. 2024
Amherst College Geology Department *Amherst, MA*

- Developed Python workflows to cross-correlate CReSIS radar depth profiles with ICESat-2 observations in Greenland and Antarctica.
- Investigated spatial variability in Greenland Ice Sheet englacial and basal structure using radar observations.
- Modeled dynamics of debris-rich glaciers using Icepack, culminating in an undergraduate honors thesis.

Student Researcher Feb.–May 2023
SIT Iceland: Climate Change and the Arctic *Iceland*

- Conducted an independent study of climate-change impacts on hydroelectric potential in glacial rivers using the CemaneigeGR4J hydrological model and Icelandic Meteorological Office data.
- Presented results in a research paper and public scientific presentation.

Summer Research Fellow Jun.–Aug. 2022
National Institute of Standards and Technology *Boulder, CO*

- Developed a Gaussian plume dispersion model in MATLAB to quantify atmospheric controls on plume geometry and performed Monte Carlo sensitivity analysis.
- Contributed to a real-time system that uses atmospheric observations to position a differential-absorption LIDAR beam for greenhouse-gas emission measurements.

Publications

Holschuh, N.; Christianson, K.; **Dienstfrey, W.**; Hills, B.; Hoffman, A. O.; Paden, J.; Winter, K.; Zuraw, R. "Entrained debris records regrowth of the Greenland Ice Sheet after the last interglacial." *Nature Geoscience* **19**, 573–580 (2026).

doi:10.1038/s41561-026-01950-1.

Stroud, J. R.; **Dienstfrey, W. J.**; Plusquellic, D. F. “Study on Local Power Plant Emissions Using Multi-Frequency Differential Absorption LIDAR and Real-Time Plume Tracking.” *Remote Sensing* **15**, 4283 (2023). doi:10.3390/rs15174283.

Preprints and Manuscripts

Krauss, Z.; Mazur, M.; Fontaine, N. K.; Ryf, R.; Neilson, D. T.; **Dienstfrey, W.**; Rose, A.; Lipovsky, B.; Denolle, M.; Abadi, S.; Hartog, R.; Wilcock, W. “Distributed acoustic sensing beyond optical repeaters offshore Central Oregon: a community dataset for a transformative technology.” Preprint (2026). doi:10.22541/essoar.15004068/v1.

Technical Reports

McFarlin, H.; **Dienstfrey, W.**; Farrell, A.; Grassi, B.; Holtkamp, S.; Murphy, N.; Nadin, E.; Parcheta, C.; Stabs, A.; West, M. “Alaska Earthquake Center Quarterly Review January–March 2025.” *Alaska Earthquake Center Reports*, No. 51 (2025). ScholarWorks.

Conference Presentations and Posters

Dienstfrey, W.; Lipovsky, B. “Imaging Ocean-Bottom Sedimentation Rates with Multi-Span Fiber Optic Sensing.” Student Seismology Workshop, oral presentation, Oct. 2026.

Dienstfrey, W.; Lipovsky, B.; Cantine, M. “Imaging Ocean-Bottom Sedimentation Rates with Multi-Span Fiber Optic Sensing.” SSA Optical Seismology Topical Meeting, poster, Oct. 2026.

Dienstfrey, W.; Lipovsky, B.; Prior-Jones, M. “A First Look at Borehole Fiber Optic Sensing Observations on Isunnguata Sermia.” IGS British Branch, oral presentation, Sep. 2026.

Holschuh, N.; **Dienstfrey, W.**; Zuraw, R. “Entrained debris as a possible record of Greenland Ice Sheet regrowth after the last interglacial period.” AGU Annual Meeting, poster, Dec. 2024.

Holschuh, N.; **Dienstfrey, W.**; Zuraw, R. “Entrained debris as a possible record of Greenland Ice Sheet regrowth after the last interglacial period.” AGU Annual Meeting, poster, Dec. 2024.

Dienstfrey, W.; Holschuh, N.; Shapero, D. “Effects of Heterogeneity in Subsurface Properties on Future Thinning of Petermann Glacier, N. Greenland.” GSA Northeastern Section Meeting, poster, Mar. 2024.

Dienstfrey, W.; Holschuh, N.; Shapero, D. “Evaluating the Role of Deep Ice Properties on Future Thinning of Petermann Glacier, N. Greenland.” AGU Annual Meeting, oral presentation, Dec. 2023.

Dienstfrey, W.; Stroud, J. R.; Plusquellic, D. F. “Monitoring Atmospheric Stability for Plume Emission Flux Measurements.” NIST Summer Research Symposium, oral presentation, Aug. 2022.

Holschuh, N.; **Dienstfrey, W.**; Nydam, T. “Is the strength of the glacier substrate expressed in fine-resolution basal morphology? Lessons from swath radar imagery and ICESat-2.” AGU Fall Meeting, poster, Dec. 2021.

Dienstfrey, W.; Holschuh, N. “Glacial Bedforms: What Information Do They Really Hold?” SURF Poster Session, poster, Aug. 2021.

Field Experience

Borehole Geophysics — Isunnguata Sermia Jul. 15–22, 2026
Cardiff University and University of Washington *Southwest Greenland*

- Joined UK-based team to hot water drill three boreholes reaching the glacier bed.
- Installed and maintained field power, fiber-optic, and data-acquisition systems during a remote glaciological field campaign.
- Collected multi-instrument geophysical observations targeting englacial hydrology, seismicity, and glacier dynamics.

Distributed Acoustic Sensing — Yellowstone Caldera Apr. 2026–Present
University of Washington *Yellowstone National Park*

- Conduct repeated DAS measurements using roadside telecommunications fiber crossing hydrothermal systems along the Yellowstone caldera rim.
- Acquire seismic observations for time-lapse tomography of hydrothermal-feature evolution.

Submarine Multi-Span Distributed Acoustic Sensing — OOI Regional Cabled Array Nov. 2025–Feb. 2026
University of Washington *Oregon*

- Conducted Multi-Span and Traditional DAS measurements on ocean-bottom fiber within the Ocean Observatories Initiative Regional Cabled Array.
- Acquired ambient seismic wavefields for surface-wave imaging of marine sediment structure.

Seismic Network Fieldwork Fall 2024–Summer 2025
Alaska Earthquake Center *Alaska*

- Serviced, diagnosed, and repaired seismic monitoring stations in remote field settings.
- Troubleshot instrumentation, telemetry, power, and data-acquisition systems supporting the Alaska regional seismic network.

Teaching Experience

Teaching Assistant 2021–2024
Amherst College Physics and Astronomy Department *Amherst, MA*

- Introductory Electromagnetism: discussion sections and help sessions (Spring 2024).
- Introductory Mechanics: grading and help sessions (Fall 2021, Fall 2023).

Peer Tutor 2021–2024
Amherst College *Amherst, MA*

- Multivariable Calculus, Introductory Electromagnetism, and Introductory Mechanics.

Technical Skills

Programming	Python, MATLAB, R, Mathematica
Geophysical methods	Distributed acoustic sensing, seismic processing, ambient-noise interferometry, surface-wave analysis, earthquake location, seismic tomography
Scientific computing	Linux, Git/GitHub, high-performance computing, parallel processing, numerical modeling, inverse methods
Geospatial	ArcGIS, Cartopy, Rasterio, photogrammetry/UAV surveying
Selected software	ObsPy, DISBA, Icepack, GENLOC
Languages	Spanish (proficient)

Honors and Awards

- UW Top Scholar Award, University of Washington, 2025.
- Magna cum laude, Amherst College, 2024.
- Rosenberg Grant, Amherst College, 2024.
- Sigma Xi qualification, Amherst College, 2024.
- Leverett-Bradley Trophy for exceptional leadership, Amherst College Rowing, 2023.

Leadership and Service

- **Captain**, Amherst College Rowing Team, 2022–2024.
- **Peer Mentor**, Spectra, Amherst College, 2021–2024.
- **Member**, Climate and Community Committee, Amherst College Physics and Astronomy, 2023–2024.